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# Clinical Study Protocol

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**Protocol Title:** A Phase 2b, Randomized, Double-Blind, Placebo-Controlled, Dose-Ranging Study to Evaluate the Efficacy and Safety of Subcutaneous Veltrazane in Adults with Moderate-to-Severe Atopic Dermatitis

**Protocol Number:** VTZ-2201 | **Version:** 3.0 (incorporating Amendment 2) | **Date:** 12 May 2026

**Sponsor:** Meridian Biopharma, Inc., 400 Harbor Gate Drive, Cambridge, MA 02141

**Investigational Product:** Veltrazane (MB-4517), a humanized IgG4 monoclonal antibody against the interleukin-31 receptor alpha subunit (IL-31RA)

**IND Number:** 165,842 | **Phase:** 2b

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## 1. Protocol Synopsis

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**Design:** Multicenter, randomized (1:1:1), double-blind, placebo-controlled, parallel-group, dose-ranging study.

**Planned enrollment:** Approximately 240 adult participants at approximately 45 sites in North America and Europe.

**Treatment arms:**

- Veltrazane 300 mg subcutaneously (SC) every 2 weeks (Q2W), following a 600 mg loading dose
- Veltrazane 300 mg SC every 4 weeks (Q4W), following a 600 mg loading dose
- Matching placebo SC Q2W

**Study duration per participant:** Up to 28 weeks total — a screening period of up to 4 weeks, a 16-week double-blind treatment period, and an 8-week safety follow-up period.

**Clinic visits:** Nine (9) scheduled clinic visits — Screening (Visit 1), Baseline/Day 1 (Visit 2), and Weeks 2, 4, 8, 12, 16 (End of Treatment), 20, and 24 (End of Study) — plus one safety telephone contact at Week 18.

**Primary endpoint:** Proportion of participants achieving EASI-75 ( $\geq 75\%$  improvement from baseline in Eczema Area and Severity Index) at Week 16.

## 2. Background and Rationale

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Atopic dermatitis (AD) is a chronic, relapsing inflammatory skin disease affecting up to 10% of adults, characterized by intense pruritus, eczematous lesions, and substantial impairment of sleep and quality of life. Despite available therapies, a significant proportion of patients with moderate-to-severe disease do not achieve adequate control with topical agents and are unwilling or unable to tolerate existing systemic options.

Interleukin-31 (IL-31) is a key mediator of pruritus in AD. Veltrazane is a humanized IgG4 monoclonal antibody that binds IL-31RA and blocks IL-31 signaling. In the completed Phase 1 program (VTZ-1101, 88 healthy volunteers) and the Phase 2a study (VTZ-2102, 96 patients with moderate-to-severe AD), Veltrazane demonstrated dose-dependent reductions in pruritus numeric rating scale (NRS) scores and EASI improvements, with an acceptable safety profile summarized in Section 9 and detailed in the current Investigator's Brochure (Edition 5).

This Phase 2b study will characterize the dose-response relationship of two Veltrazane regimens to support dose selection for Phase 3.

## 3. Objectives and Endpoints

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### 3.1 Primary Objective

To evaluate the efficacy of two dose regimens of Veltrazane compared with placebo in adults with moderate-to-severe AD, as measured by the proportion of participants achieving EASI-75 at Week 16.

### 3.2 Secondary Objectives

- Proportion of participants achieving an Investigator's Global Assessment (IGA) score of 0 or 1 with a  $\geq 2$ -point reduction from baseline at Week 16
- Change from baseline in weekly average Peak Pruritus NRS at Weeks 4 and 16
- Proportion of participants with a  $\geq 4$ -point improvement in Peak Pruritus NRS at Week 16
- Change from baseline in Dermatology Life Quality Index (DLQI) at Week 16
- Safety and tolerability of Veltrazane (adverse events, laboratory assessments, vital signs, ECGs, immunogenicity)

### 3.3 Exploratory Objectives

Pharmacokinetics of Veltrazane; serum biomarkers of type 2 inflammation; patient-reported sleep disturbance.

## 4. Study Design

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This is a randomized, double-blind, placebo-controlled, parallel-group study. Eligible participants will be randomized 1:1:1, stratified by baseline disease severity (IGA 3 vs. 4) and geographic region. All

participants, investigators, site personnel, and sponsor staff involved in study conduct will remain blinded to treatment assignment until database lock.

Participants who complete the Week 16 assessment enter an 8-week off-treatment safety follow-up. An independent Data and Safety Monitoring Board (DSMB) will review unblinded safety data approximately every 12 weeks.

## **5. Study Population**

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### **5.1 Key Inclusion Criteria**

1. Male or female, 18 to 75 years of age at screening.
2. Chronic atopic dermatitis (per Hanifin and Rajka criteria) diagnosed at least 1 year before screening.
3. EASI score  $\geq 16$ , IGA score  $\geq 3$ , and  $\geq 10\%$  body surface area involvement at screening and baseline.
4. Documented history, within 6 months before screening, of inadequate response to a medium-or-higher-potency topical corticosteroid regimen, or documented medical contraindication to topical therapy.
5. Willing and able to comply with clinic visits, the daily electronic pruritus diary, and study-related procedures.
6. Female participants of childbearing potential must use a highly effective method of contraception from screening through 16 weeks after the last dose.

### **5.2 Key Exclusion Criteria**

1. Active skin infection requiring systemic treatment within 2 weeks of baseline, or history of recurrent serious infections.
2. Positive or indeterminate interferon-gamma release assay for tuberculosis at screening, unless documented completed prophylaxis.
3. Chronic hepatitis B or C infection, or HIV infection.
4. Receipt of any live (attenuated) vaccine within 12 weeks before baseline or planned during the study.
5. Treatment with dupilumab or any other biologic therapy within 12 weeks or 5 half-lives (whichever is longer) before baseline.
6. Systemic immunosuppressants (e.g., cyclosporine, methotrexate, JAK inhibitors) within 4 weeks before baseline.
7. Pregnant or breastfeeding, or planning pregnancy during the study period.
8. History of malignancy within 5 years, except adequately treated non-melanoma skin cancer or cervical carcinoma in situ.
9. ALT or AST  $> 2 \times$  the upper limit of normal, or eGFR  $< 45$  mL/min/1.73 m<sup>2</sup> at screening.

## **6. Study Treatments**

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### **6.1 Investigational Product**

Veltrazane 150 mg/mL solution for subcutaneous injection in single-use prefilled syringes. Loading dose of 600 mg (two injections) at Baseline, then 300 mg per the assigned regimen. Placebo syringes are identical in appearance.

Injections are administered by trained site personnel at clinic visits; the Q4W arm receives placebo injections at alternating Q2W visits to maintain the blind. Injection sites (abdomen, thigh, upper arm) are rotated.

## 6.2 Concomitant Therapy

**Permitted:** bland emollients (required, at least twice daily); low-potency topical corticosteroids as rescue for intolerable flares, with use recorded in the diary; oral antihistamines at stable doses.

**Prohibited:** systemic corticosteroids, other biologics, systemic immunosuppressants, phototherapy, and live vaccines. Use of prohibited rescue therapy results in discontinuation of study treatment but not of study participation.

## 7. Schedule of Activities

| Assessment                   | Scr<br>(V1) | BL/D1<br>(V2)                                      | W2<br>(V3) | W4<br>(V4) | W8<br>(V5) | W12<br>(V6) | W16/EOT<br>(V7) | W18 | W20<br>(V8) | W24/EOS<br>(V9) |
|------------------------------|-------------|----------------------------------------------------|------------|------------|------------|-------------|-----------------|-----|-------------|-----------------|
| Informed consent             | X           |                                                    |            |            |            |             |                 |     |             |                 |
| Eligibility review           | X           | X                                                  |            |            |            |             |                 |     |             |                 |
| Physical exam, vital signs   | X           | X                                                  | X          | X          | X          | X           | X               |     | X           | X               |
| EASI, IGA, BSA               | X           | X                                                  | X          | X          | X          | X           | X               |     | X           | X               |
| Pruritus NRS (daily e-diary) | X           | -----<br>continuous<br>through<br>Week 24<br>----- |            |            |            |             |                 |     |             |                 |
| DLQI                         |             | X                                                  |            | X          | X          | X           | X               |     |             | X               |
| 12-lead ECG                  | X           |                                                    |            |            |            |             | X               |     |             |                 |
| Safety laboratory panel      | X           | X                                                  | X          | X          | X          | X           | X               |     | X           | X               |
| Blood sampling volume        | ~20 mL      | ~15 mL                                             | ~15 mL     | ~15 mL     | ~15 mL     | ~15 mL      | ~15 mL          |     | ~15 mL      | ~15 mL          |
| Serum pregnancy test (WOCBP) | X           |                                                    |            |            |            |             |                 |     |             |                 |
| Urine pregnancy test (WOCBP) |             | X                                                  |            | X          |            | X           | X               |     |             | X               |
| PK / immunogenicity          |             | X                                                  | X          | X          |            | X           | X               |     |             | X               |

| Assessment                | Scr (V1) | BL/D1 (V2) | W2 (V3) | W4 (V4) | W8 (V5) | W12 (V6) | W16/EOT (V7) | W18 | W20 (V8) | W24/EOS (V9) |
|---------------------------|----------|------------|---------|---------|---------|----------|--------------|-----|----------|--------------|
| sample                    |          |            |         |         |         |          |              |     |          |              |
| Study drug administration |          | X          | X       | X       | X       | X        |              |     |          |              |
| Telephone safety contact  |          |            |         |         |         |          |              | X   |          |              |
| Adverse event review      | X        | X          | X       | X       | X       | X        | X            | X   | X        | X            |

Total blood volume over the study will not exceed approximately 150 mL, which is less than a standard blood donation.

## 8. Discontinuation and Withdrawal

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Participants may withdraw consent and discontinue at any time without penalty and without affecting their regular medical care. The investigator must discontinue study treatment for: pregnancy; anaphylaxis or other serious hypersensitivity reaction related to study drug; ALT or AST >5× ULN; any serious infection requiring prolonged systemic therapy; or at the participant's request. Participants who discontinue treatment early will be asked to complete early-termination assessments and the safety follow-up.

## 9. Safety Considerations

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### 9.1 Identified Risks of Veltrazane

Based on the completed Phase 1 and Phase 2a studies (n=184 exposed):

- **Injection-site reactions** (erythema, swelling, pain) — approximately **18%** of participants; predominantly mild, resolving within 48 hours.
- **Conjunctivitis** (allergic or unspecified) — approximately **9%**, versus 2% on placebo; generally mild-to-moderate and manageable with lubricating or anti-inflammatory eye drops. Participants with persistent or worsening eye symptoms must be referred for ophthalmologic evaluation.
- **Headache** — approximately **11%**, mild and transient.
- **Transient eosinophilia** ( $>1.5 \times 10^9/\text{L}$  without clinical sequelae) — approximately **4%**; monitored via the safety laboratory panel at every visit.

### 9.2 Potential Risks

- **Serious hypersensitivity, including anaphylaxis** — one non-fatal anaphylactic reaction was reported in the Phase 2a study (0.5% of exposed participants), occurring within 30 minutes of injection. All participants must be observed in clinic for at least 60 minutes after each of the first three injections and 30 minutes thereafter. Sites must have anaphylaxis management capability available.

- **Serious infections** — as an immunomodulator, Veltrazane may theoretically increase infection risk; no imbalance has been observed to date. Participants must be counseled to report fever or signs of infection promptly.
- **Unknown risks**, including risks to a fetus. Effects of Veltrazane on pregnancy are unknown; contraception requirements apply (Section 5.1).

Risks of study procedures include bruising, discomfort, or rarely infection at venipuncture sites, and skin irritation from ECG electrodes.

### 9.3 Adverse Event Reporting

All adverse events are recorded from consent through the End of Study visit. Serious adverse events must be reported to the sponsor within 24 hours of site awareness and to the IRB/IEC per local requirements. The sponsor will submit expedited safety reports (IND safety reports) to FDA per 21 CFR 312.32.

## 10. Statistical Considerations

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Approximately 240 participants (80 per arm) provide  $\geq 90\%$  power to detect a 25-percentage-point difference in EASI-75 response versus placebo at a two-sided alpha of 0.05, assuming a placebo response of 15% and 10% dropout. The primary analysis uses a Cochran–Mantel–Haenszel test stratified by baseline IGA and region on the intent-to-treat population, with non-responder imputation for missing Week 16 data. Continuous endpoints use a mixed model for repeated measures (MMRM).

## 11. Ethics and Regulatory Considerations

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### 11.1 Ethical Conduct

The study will be conducted in accordance with the protocol, ICH E6(R2) Good Clinical Practice, the Declaration of Helsinki, and applicable regulations including 21 CFR Parts 50, 56, and 312.

### 11.2 Informed Consent

Written informed consent must be obtained by the investigator or qualified designee before any study-specific procedure. The consent form must include all elements required by 21 CFR 50.25, must accurately describe the risks documented in Section 9, and must be approved by the reviewing IRB/IEC. Participants must receive a signed copy.

### 11.3 Confidentiality

Participants are identified by a coded study number. Direct access to identifiable records is limited to the investigator and site staff, the sponsor's authorized monitors and auditors, the reviewing IRB/IEC, and regulatory authorities, including the U.S. Food and Drug Administration, which may inspect study records. Data are processed in access-controlled, validated systems in accordance with applicable data-protection law.

## **11.4 Care and Compensation for Research-Related Injury**

If a participant is injured as a direct result of study participation or study drug administration, the sponsor, Meridian Biopharma, will pay the reasonable and necessary costs of diagnosis and medical treatment of the injury, to the extent those costs are not covered by the participant's health insurance or other third-party coverage. No other form of compensation (for example, for lost wages, disability, or discomfort) will be provided. The informed consent form must explain both the availability of medical treatment and this compensation position, and must identify where participants may obtain further information. Payment for participation (visit stipends) is separate and described in the consent form.

## **11.5 Voluntary Participation**

Participation is voluntary. Refusal to participate or withdrawal at any time involves no penalty or loss of benefits to which the participant is otherwise entitled.

## **12. Study Administration**

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**Coordinating Investigator:** Dr. Elena Vasquez, MD, PhD — Meridian Clinical Research Center, Boston, MA

**Sponsor Medical Monitor:** Dr. James Okafor, MD — +1 (555) 310-4471 (24-hour medical emergency line)

**Central IRB:** Beacon Hill Independent Review Board — +1 (555) 310-9920

Questions regarding participant rights are directed to the reviewing IRB. Suspected study-related injuries must be reported to the site investigator and the sponsor medical monitor.